

SWAVIK INTERNSHIP – 2026

**PROJECT TITLE:
ICU Patient Deterioration Early Warning System**



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ABSTRACT

The **ICU Patient Deterioration Early Warning System** is a real-time clinical monitoring application developed to identify critically ill patients at risk of rapid deterioration. The system simulates ICU patient vital signs, including heart rate, oxygen saturation, respiratory rate, blood pressure, and body temperature. It validates and cleans incoming data before calculating the **National Early Warning Score 2 (NEWS2)** for each patient. Based on the calculated score, patients are classified into Low, Medium, High, or Critical risk categories. An intelligent alert generation mechanism notifies healthcare professionals when a patient's condition worsens while minimizing unnecessary repeated alerts through alert suppression. All clinical alerts are securely stored in an SQLite database for auditing and future analysis. The application provides an interactive dashboard developed using Streamlit and Plotly to visualize patient risk levels, trends, and system performance. The proposed system enhances clinical decision-making by enabling early detection of patient deterioration. It demonstrates how real-time analytics and visualization can improve ICU monitoring and patient safety. This project serves as an effective academic prototype for intelligent healthcare monitoring and early warning systems.

INTRODUCTION

The **Intensive Care Unit (ICU)** is a specialized hospital department that provides continuous monitoring and treatment for critically ill patients. Patients admitted to the ICU often experience rapid changes in their health condition, making early detection of deterioration essential for improving survival rates and reducing medical complications. Healthcare professionals regularly monitor vital signs such as heart rate, respiratory rate, blood pressure, oxygen saturation (SpO₂), body temperature, and level of consciousness to assess a patient's condition. However, continuously monitoring multiple patients manually can be difficult and time-consuming, increasing the possibility of delayed clinical intervention.

Advancements in healthcare technology have enabled hospitals to adopt digital monitoring systems that collect patient data in real time. Intelligent clinical monitoring systems analyze this data automatically and assist medical staff by identifying abnormal physiological conditions at an early stage. One of the most widely accepted methods for assessing patient deterioration is the **National Early Warning Score 2 (NEWS2)**. NEWS2 evaluates multiple vital parameters and assigns a score based on their severity. The total score helps classify patients into different risk categories, allowing healthcare professionals to prioritize treatment for patients requiring immediate attention.

The **ICU Patient Deterioration Early Warning System** developed in this project is designed to simulate real-time monitoring of ICU patients and automatically identify individuals at risk of clinical deterioration. The system generates realistic patient vital signs, validates the collected data, handles missing or invalid values, and calculates the NEWS2 score for each patient. Based on the calculated score, patients are classified into **Low, Medium, High, and Critical** risk categories. High-risk patients are immediately identified, enabling faster clinical decision-making and timely medical intervention.

To improve the reliability of monitoring, the system incorporates an intelligent alert generation mechanism that notifies healthcare professionals whenever a patient's condition worsens. It also implements an alert suppression technique to prevent repeated notifications for the same patient within a short period, thereby reducing alarm fatigue among clinicians. All generated alerts are securely stored in an SQLite database, providing a complete audit trail that supports future analysis and system evaluation.

The application is developed using **Python, Streamlit, Pandas, NumPy, Plotly, and SQLite**, providing an interactive dashboard for visualizing patient conditions, risk distribution, alert history, and clinical trends. Although the project uses simulated patient data for academic purposes, it closely represents the workflow of modern intelligent healthcare monitoring systems. Overall, the proposed system demonstrates how real-time data analytics, automated NEWS2 scoring, and interactive visualization can enhance patient safety, support clinical decision-making, and improve the efficiency of ICU monitoring.

LITERATURE SURVEY

A literature survey helps in understanding the existing research, methodologies, and technologies related to patient monitoring and early warning systems in intensive care environments. Several studies have focused on using clinical scoring systems, machine learning, and real-time monitoring technologies to identify patient deterioration at an early stage. These studies emphasize that timely detection of abnormal physiological conditions can significantly reduce mortality, improve patient safety, and support better clinical decision-making. The following literature survey reviews research papers that are closely related to the proposed **ICU Patient Deterioration Early Warning System**.

1. Royal College of Physicians (2017). National Early Warning Score (NEWS2): Standardising the Assessment of Acute Illness Severity in the NHS.

The Royal College of Physicians introduced the National Early Warning Score 2 (NEWS2) as a standardized clinical scoring system for assessing patient deterioration. NEWS2 evaluates six physiological parameters, including respiratory rate, oxygen saturation, systolic blood pressure, pulse rate, body temperature, and level of consciousness. The study demonstrates that standardized scoring enables healthcare professionals to identify critically ill patients earlier and initiate timely medical intervention. NEWS2 has become one of the most widely adopted early warning systems in hospitals worldwide. This research forms the foundation of the proposed project because the system calculates NEWS2 scores to classify patient risk levels.

2. Subbe, C. P., et al. (2001). Validation of a Modified Early Warning Score in Medical Admissions.

This research evaluates the effectiveness of the Modified Early Warning Score (MEWS) in predicting patient deterioration. The study found that increasing physiological abnormalities are strongly associated with higher risks of intensive care admission and mortality. The authors conclude that early warning scores can assist clinicians in recognizing deteriorating patients before severe complications occur. The findings support the importance of automated monitoring systems that continuously calculate warning scores, similar to the approach implemented in the proposed project.

3. Escobar, G. J., et al. (2020). Automated Identification of Adults at Risk for Clinical Deterioration Using Electronic Health Records.

This study presents an automated clinical decision support system that analyzes electronic health records to predict patient deterioration. Machine learning algorithms process patient vital signs and laboratory information to identify individuals at high risk. The research demonstrates that automated monitoring improves prediction accuracy and enables faster clinical response compared to manual observation alone. The proposed ICU Patient Deterioration Early Warning System follows a similar concept by automatically analyzing patient vital signs and generating clinical alerts based on NEWS2 scoring.

4. Clifton, D. A., et al. (2014). Predictive Monitoring of Mobile Patients by Combining Clinical Observations with Data Analytics.

The authors investigate the use of continuous physiological monitoring and data analytics for predicting patient deterioration. The study emphasizes that real-time monitoring systems are capable of detecting subtle physiological changes that may not be immediately recognized during routine clinical examinations. The research highlights the importance of integrating intelligent algorithms with patient monitoring systems to improve patient safety. This concept is reflected in the proposed system through continuous patient monitoring, automated risk classification, and intelligent alert generation.

5. Sendak, M. P., et al. (2020). A Path for Translation of Machine Learning Products into Healthcare Delivery.

This paper discusses the implementation of machine learning and artificial intelligence applications in clinical practice. The authors explain that healthcare decision-support systems should provide reliable predictions while maintaining transparency and usability for clinicians. The study also emphasizes the importance of data validation, audit logging, and user-friendly dashboards. These recommendations are incorporated into the proposed project through patient data validation, SQLite audit logging, and an interactive Streamlit dashboard for monitoring ICU patients.

OBJECTIVES

The primary objective of the **ICU Patient Deterioration Early Warning System** is to develop an intelligent clinical monitoring application that can continuously analyze patient vital signs and identify patients at risk of clinical deterioration. The specific objectives of the project are as follows:

1. To develop a real-time ICU patient monitoring system that simulates and continuously monitors patient vital signs.
2. To calculate the **National Early Warning Score 2 (NEWS2)** using physiological parameters such as heart rate, respiratory rate, oxygen saturation, blood pressure, body temperature, and level of consciousness.
3. To classify patients into **Low, Medium, High, and Critical** risk categories based on their NEWS2 scores for effective clinical prioritization.
4. To generate automated clinical alerts whenever a patient's condition deteriorates, enabling timely medical intervention.
5. To implement data validation and missing-value imputation techniques to improve the accuracy and reliability of patient monitoring.
6. To reduce unnecessary repeated notifications through an intelligent alert suppression mechanism, thereby minimizing alarm fatigue among healthcare professionals.
7. To maintain a secure audit log of all generated alerts using an SQLite database for future analysis and clinical record management.
8. To develop an interactive Streamlit dashboard with graphical visualizations using Plotly for monitoring patient status, risk distribution, and historical trends.
9. To demonstrate the practical application of Python, data analytics, and healthcare technologies in developing an intelligent early warning system for intensive care units.
10. To improve patient safety and support faster clinical decision-making through continuous monitoring and early detection of patient deterioration.

PROJECT SCOPE AND OVERVIEW

The **ICU Patient Deterioration Early Warning System** is designed to support healthcare professionals by continuously monitoring patient vital signs and identifying patients who are at risk of clinical deterioration. The system focuses on providing an intelligent and automated approach to patient monitoring by analyzing physiological parameters and calculating the **National Early Warning Score 2 (NEWS2)** in real time. By detecting abnormal patient conditions at an early stage, the system enables timely medical intervention, improves patient safety, and assists clinicians in making informed decisions.

The scope of the project includes the simulation of ICU patient data representing vital signs such as heart rate, respiratory rate, oxygen saturation (SpO₂), systolic blood pressure, body temperature, oxygen therapy status, and level of consciousness. The generated patient data undergoes validation to identify invalid or missing values, followed by appropriate preprocessing to ensure accurate clinical analysis. The system then calculates the NEWS2 score for every patient and classifies patients into **Low, Medium, High, and Critical** risk categories according to their physiological condition.

An intelligent alert generation mechanism is incorporated to notify healthcare professionals whenever a patient's condition deteriorates beyond predefined clinical thresholds. To minimize alarm fatigue, the application implements an alert suppression strategy that prevents repeated notifications for the same patient within a specified time interval unless the patient's condition worsens further. All generated alerts are securely stored in an SQLite database, creating a complete audit trail for future analysis, reporting, and system evaluation.

The project also includes an interactive dashboard developed using **Streamlit**, where healthcare professionals can monitor patient status through real-time charts, heat maps, patient risk rankings, and historical alert records. Visualization using **Plotly** enables quick interpretation of patient conditions and supports efficient clinical decision-making.

Although the proposed system uses simulated patient data for academic purposes, its architecture closely resembles modern hospital monitoring systems that integrate real-time analytics, automated clinical scoring, and intelligent alert management. The project demonstrates the practical application of Python programming, healthcare data analytics, and visualization technologies in developing an effective early warning system for intensive care units.

TECHNICAL STACK

The **ICU Patient Deterioration Early Warning System** is developed using modern programming languages, frameworks, libraries, and database technologies that support real-time patient monitoring, data analysis, visualization, and alert management. The technical stack used in the project is presented in Table 1.1.

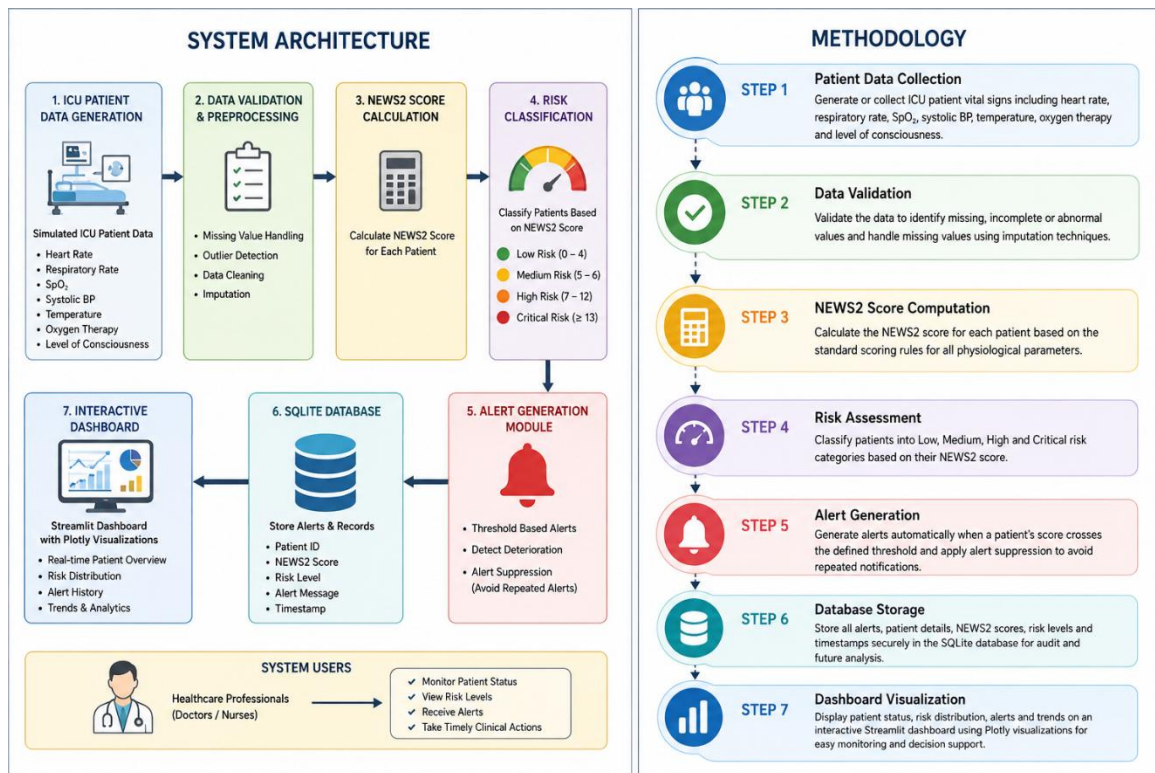
Table 1.1: Technical Stack Used in the Project

Sl. No.	Technology / Tool	Category	Purpose in the Project
1	Python	Programming Language	Used to develop the complete application and implement business logic.
2	Streamlit	Web Application Framework	Provides an interactive dashboard for monitoring ICU patients and displaying results.
3	Pandas	Data Processing Library	Used for handling, cleaning, validating, and processing patient datasets.
4	NumPy	Numerical Computing Library	Performs mathematical calculations and supports NEWS2 score computation.
5	Plotly	Data Visualization Library	Creates interactive charts, graphs, and patient risk visualizations.
6	SQLite	Database Management System	Stores patient alerts, audit logs, and historical monitoring records.
7	Datetime	Python Module	Manages timestamps for alerts, patient records, and historical analysis.
8	Random	Python Module	Generates simulated ICU patient vital signs for testing and demonstration.
9	HTML/CSS	Frontend Technologies	Enhances the appearance and layout of the Streamlit application.
10	Visual Studio Code	Development Environment	Used for coding, debugging, and testing the application.
11	pip	Package Manager	Installs and manages the required Python libraries and dependencies.

12	Windows System	Operating Execution Environment	Provides the platform for developing and running the application.
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The technical stack selected for this project provides a reliable and efficient environment for implementing a real-time ICU patient monitoring application. **Python** serves as the core programming language due to its simplicity and extensive support for healthcare data analysis. **Streamlit** enables the development of an interactive web-based dashboard without requiring complex frontend development. **Pandas** and **NumPy** perform data preprocessing, validation, and numerical computations necessary for calculating NEWS2 scores. **Plotly** offers dynamic visualizations that help healthcare professionals quickly understand patient conditions through charts and heat maps. **SQLite** maintains a secure database of generated alerts and audit logs, while supporting modules such as **Datetime** and **Random** manage timestamps and simulated patient data generation. Together, these technologies provide an integrated platform for intelligent patient monitoring, early warning detection, and clinical decision support.

SYSTEM ARCHITECTURE & METHODOLOGY



SYSTEM ARCHITECTURE

The **ICU Patient Deterioration Early Warning System** is designed as a real-time patient monitoring application that continuously analyzes patient vital signs, calculates the National Early Warning Score 2 (NEWS2), identifies patients at risk, and generates timely clinical alerts. The system integrates data processing, risk assessment, alert management, database storage, and visualization into a single workflow. The architecture ensures efficient monitoring of ICU patients while supporting healthcare professionals in making faster and more accurate clinical decisions.

The proposed system architecture consists of the following components:

1. ICU Patient Data Generation

The first component of the system generates or receives patient physiological data. For this project, simulated ICU patient data is created for approximately 200 patients. The generated data includes heart rate, respiratory rate, oxygen saturation (SpO₂), systolic blood pressure, body temperature, oxygen therapy status, and level of consciousness. These parameters represent the vital signs required for NEWS2 score calculation.

2. Data Validation and Preprocessing

The collected patient data is validated to identify missing, incomplete, or abnormal values. Invalid observations are corrected using preprocessing techniques and missing values are imputed whenever possible. This step improves data quality and ensures that only reliable information is used for clinical analysis.

3. NEWS2 Score Calculation

After preprocessing, the system calculates the National Early Warning Score 2 (NEWS2) for every patient based on standard clinical scoring rules. Each physiological parameter contributes to the overall score according to predefined thresholds. The calculated score represents the severity of the patient's clinical condition.

4. Risk Classification

Based on the NEWS2 score, patients are automatically classified into four risk categories: **Low Risk**, **Medium Risk**, **High Risk**, and **Critical Risk**. This classification helps healthcare professionals prioritize patients who require immediate medical attention.

5. Alert Generation Module

Whenever a patient's NEWS2 score exceeds predefined clinical limits, the system automatically generates an alert. An alert suppression mechanism prevents repeated notifications for the same patient within a specified time period, thereby reducing alarm fatigue while ensuring that significant changes in patient condition continue to trigger alerts.

6. SQLite Database

All generated alerts and patient monitoring records are stored securely in an SQLite database. The database maintains an audit history containing patient identifiers, NEWS2 scores, alert messages, timestamps, and risk levels. This information supports reporting, historical analysis, and future system evaluation.

7. Interactive Dashboard

The processed information is displayed through an interactive Streamlit dashboard. The dashboard provides patient risk rankings, heat maps, alert summaries, historical trends, and graphical visualizations using Plotly. These visual tools enable clinicians to monitor ICU patients efficiently and make timely clinical decisions.

METHODOLOGY

The methodology adopted for the proposed system follows a systematic approach for monitoring ICU patients and detecting early signs of clinical deterioration.

Step 1: Patient Data Collection

The system begins by generating simulated ICU patient data containing important physiological parameters required for patient assessment. This simulated dataset represents real-time monitoring data collected from ICU medical devices.

Step 2: Data Validation

The generated patient information is validated to detect missing or abnormal values. Invalid records are corrected or replaced using appropriate preprocessing techniques to improve data reliability.

Step 3: NEWS2 Score Computation

The validated physiological parameters are analyzed using the NEWS2 clinical scoring algorithm. Individual parameter scores are calculated and combined to obtain the total NEWS2 score for each patient.

Step 4: Risk Assessment

Based on the NEWS2 score, each patient is classified into Low, Medium, High, or Critical risk categories. Patients with higher scores receive higher clinical priority.

Step 5: Alert Generation

The system continuously monitors patient scores. Whenever deterioration is detected, automated alerts are generated and duplicate notifications are minimized through an alert suppression mechanism.

Step 6: Database Storage

All patient alerts, timestamps, NEWS2 scores, and clinical information are stored in an SQLite database to maintain a complete audit trail for future analysis.

Step 7: Dashboard Visualization

Finally, the processed information is displayed on the Streamlit dashboard using interactive charts, heat maps, patient rankings, and alert summaries. These visualizations help healthcare professionals quickly identify high-risk patients and improve clinical decision-making.

The proposed methodology ensures accurate patient monitoring, timely identification of deterioration, efficient alert management, secure record maintenance, and effective visualization of patient health information, making it suitable for intelligent ICU monitoring applications.

KEY IMPLEMENTATION FEATURES

The **ICU Patient Deterioration Early Warning System** is implemented using Python and modern data analysis technologies to provide intelligent monitoring of ICU patients. The system integrates data processing, clinical scoring, alert generation, database management, and visualization into a single application. The major implementation features of the proposed system are described below.

1. ICU Patient Data Simulation

The application generates realistic physiological data for approximately 200 ICU patients. The simulated dataset includes heart rate, respiratory rate, oxygen saturation (SpO₂), systolic blood pressure, body temperature, oxygen therapy status, and level of consciousness. This simulated environment enables testing and demonstration without using sensitive patient information.

2. Data Validation and Preprocessing

Before performing clinical analysis, the system validates the patient data to identify missing, incomplete, or abnormal values. Missing values are appropriately handled using imputation techniques, while invalid readings are corrected or removed. This preprocessing step improves the reliability and accuracy of patient monitoring.

3. NEWS2 Score Calculation

The system automatically calculates the **National Early Warning Score 2 (NEWS2)** for every patient based on multiple physiological parameters. Each parameter contributes to the overall score according to standard clinical guidelines, allowing accurate assessment of patient condition.

4. Patient Risk Classification

Based on the calculated NEWS2 score, patients are automatically classified into four categories: **Low Risk, Medium Risk, High Risk, and Critical Risk**. This classification helps healthcare professionals prioritize patients who require immediate medical attention.

5. Intelligent Alert Generation

Whenever a patient's NEWS2 score exceeds predefined clinical thresholds, the application generates an automatic clinical alert. The alert mechanism continuously monitors patient conditions and immediately notifies healthcare professionals about deteriorating patients.

6. Alert Suppression Mechanism

To minimize alarm fatigue, the system implements an alert suppression feature that prevents repeated notifications for the same patient within a specified time interval unless the patient's condition worsens significantly. This improves the effectiveness of clinical alerts.

7. SQLite Audit Logging

All generated alerts, patient identifiers, NEWS2 scores, timestamps, and risk categories are securely stored in an SQLite database. The audit log provides historical records for future analysis, reporting, and system evaluation.

8. Interactive Dashboard

The application provides a user-friendly dashboard developed using **Streamlit**. Healthcare professionals can monitor patient conditions, view alert summaries, analyze historical records, and access real-time patient information through an intuitive interface.

9. Data Visualization

The system uses **Plotly** to generate interactive charts, heat maps, patient risk distributions, and trend graphs. These visualizations help clinicians quickly identify high-risk patients and understand the overall condition of the ICU.

10. Real-Time Clinical Decision Support

The integration of automated NEWS2 scoring, intelligent alert generation, audit logging, and interactive visualization provides effective clinical decision support. The system enables early identification of patient deterioration, allowing healthcare professionals to take timely medical action and improve patient safety.

CONCLUSION

The **ICU Patient Deterioration Early Warning System** successfully demonstrates the development of an intelligent patient monitoring application that supports the early identification of critically ill patients. The system continuously analyzes patient vital signs, validates the collected data, calculates the **National Early Warning Score 2 (NEWS2)**, and classifies patients into different risk categories based on their physiological condition. By automatically detecting patient deterioration and generating timely clinical alerts, the system assists healthcare professionals in making faster and more informed medical decisions.

The project integrates modern technologies such as **Python, Streamlit, Pandas, NumPy, Plotly**, and **SQLite** to provide an interactive and efficient monitoring platform. Features such as automated NEWS2 scoring, intelligent alert generation, alert suppression, audit logging, and graphical visualization improve the overall effectiveness of patient monitoring while reducing unnecessary alarm fatigue. The interactive dashboard enables healthcare professionals to monitor patient status, view historical trends, and identify high-risk patients through intuitive visualizations.

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